

Class 12

David Majeed (A17885958)

```
#Section 1. Protion of G/G in Population
mxl<-read.csv("373531-SampleGenotypes-Homo_sapiens_Variation_Sample_rs8067378.csv")
head(mxl)
```

	Sample..Male.Female.Unknown.	Genotype..forward.strand.	Population.s.	Father
1	NA19648 (F)	A A	ALL, AMR, MXL	-
2	NA19649 (M)	G G	ALL, AMR, MXL	-
3	NA19651 (F)	A A	ALL, AMR, MXL	-
4	NA19652 (M)	G G	ALL, AMR, MXL	-
5	NA19654 (F)	G G	ALL, AMR, MXL	-
6	NA19655 (M)	A G	ALL, AMR, MXL	-
Mother				
1	-			
2	-			
3	-			
4	-			
5	-			
6	-			

```
table(mxl$Genotype..forward.strand.)/nrow(mxl)
```

A A	A G	G A	G G
0.343750	0.328125	0.187500	0.140625

Section 4

Q.13 +14

```
expr<-read.table("rs8067378_ENSG00000172057.6.txt")
#Lets import our data
head(expr)
```

```
      sample geno      exp
1 HG00367   A/G 28.96038
2 NA20768   A/G 20.24449
3 HG00361   A/A 31.32628
4 HG00135   A/A 34.11169
5 NA18870   G/G 18.25141
6 NA11993   A/A 32.89721
```

```
nrow(expr)
```

```
[1] 462
```

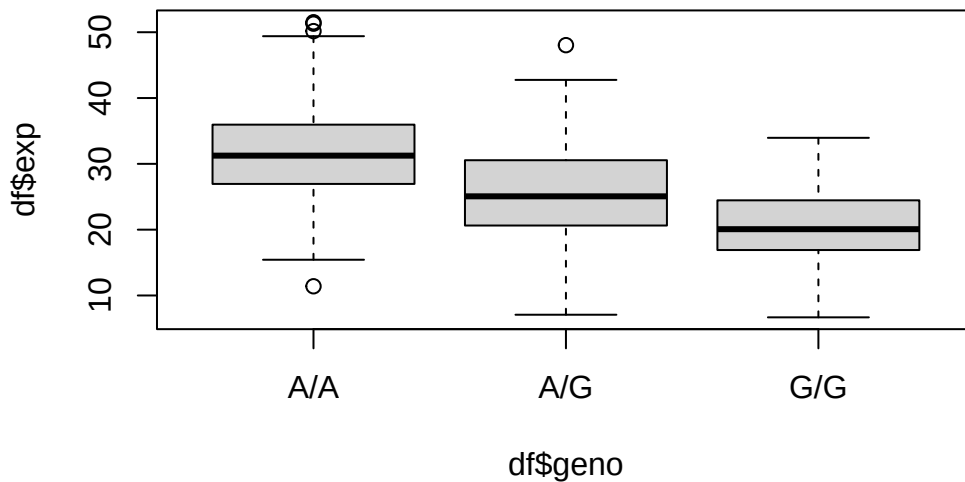
```
table(expr$geno)
```

```
A/A A/G G/G
108 233 121
```

```
#Lets get some basic understanding of our data by looking at the header, number of rows, and
df<- data.frame(expr)
```

```
#To further examine, lets get our data into a data frame
```

```
bp_out<-boxplot(df$exp ~ df$geno, data=df)
```



#Lets use the boxplot function on our data frame to get some stats like the median. It will o

bp_out\$stats

	[,1]	[,2]	[,3]
[1,]	15.42908	7.07505	6.67482
[2,]	26.95022	20.62572	16.90256
[3,]	31.24847	25.06486	20.07363
[4,]	35.95503	30.55183	24.45672
[5,]	49.39612	42.75662	33.95602

#Q.13 There are 108 A/A,233 A/G, and 121 G/G. The medians, respectively are, 31.25, 25.06, and

```
library(ggplot2)
```

```
#Load up ggplot
```

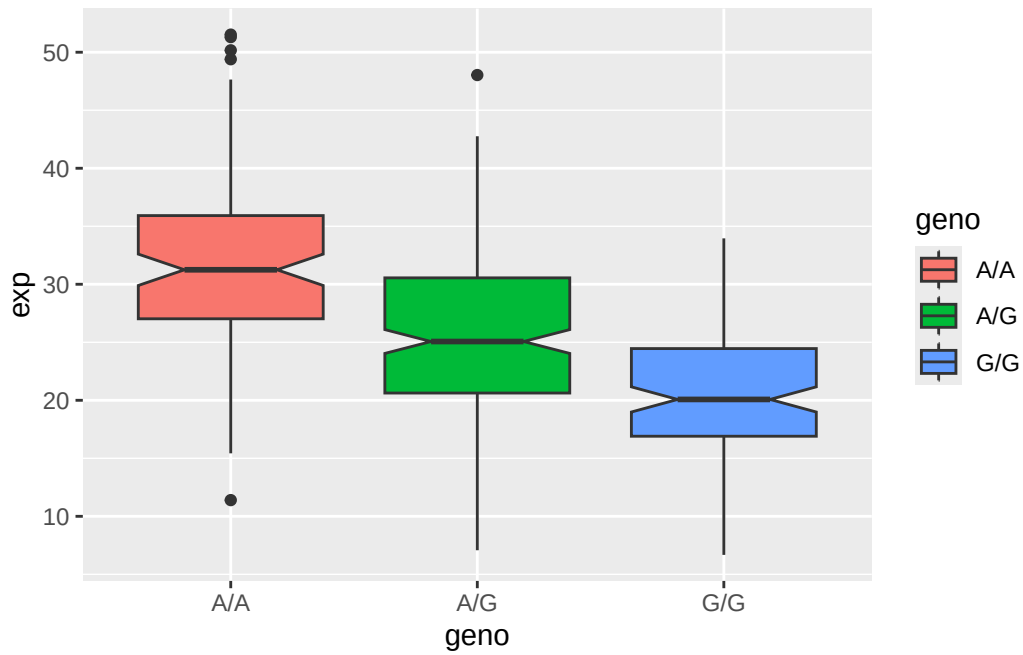
```
ggplot(expr)+
```

```
#ggplot allows us to use the plot function with (expr) highlighting the use of the expr data
```

```
  aes(geno, exp, fill=geno)+
```

```
  #aes helps us determine the x and y axis as the genotype and expression levels. fill=geno
```

```
  geom_boxplot(notch = T)
```



`#geom_boxplot` allows us to specify that we are making a box plot graph and `notch` gives it the

`#Q.14` From the boxplot we see that homozygous A has higher expression levels in comparison to